

Generative Clinical Narrative and Risk Explanation from ICU Time-Series Data Using Large Language Models

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The Problem

ICU clinicians face a data overload problem

- High-frequency vitals, irregular labs, ventilation settings, clinical notes — stored in relational tables
- Risk scores give a number but not **why** or **what to do next**
- Raw tables are hard to scan under time pressure at the bedside

Gap: Transform structured ICU time-series into **interpretable, point-in-time** natural-language explanations — without leaking future data.

Objectives

#	Objective
1	Load real ICU stays from eICU-CRD demo (PhysioNet)
2	Build a multi-agent pipeline for vitals, labs, risk, and narrative
3	Enforce point-in-time snapshots — all views reflect data known only up to selected ICU hour
4	Deliver Critical Patient Brief — SBAR, 6hr trends, explainable alerts, actions
5	Support clinician dialogue + feedback overlay (human-in-the-loop)
6	Orchestrate batch runs via Apache Airflow with persisted artifacts

Scope: Research prototype — not validated for clinical deployment

Dataset & Data Pipeline

eICU Collaborative Research Database — Demo v2.0.1

- 2,500+ de-identified ICU stays · 20 hospitals · open PhysioNet access
- Local store: `data/eicu-crd-demo/` (SQLite + CSV.gz)
- Loader: `eicu_loader.py` → unified **PatientState** schema (Pydantic)

Dual mode

- `ICU_DATA_SOURCE=eicu` — real measurements (primary)
- `ICU_DATA_SOURCE=synthetic` — fallback when charting is sparse

Mapped modalities: vitals · labs · notes · respiratory charting · diagnosis

System Architecture

ICU Multi-Agent System Architecture



Core Contribution — Point-in-Time Snapshots

Problem: Future-data leakage invalidates bedside decision support

```
eICU SQLite → build_timeline() → snapshot_at_hour(h) → all downstream inference
```

Function	Role
build_timeline()	Merge hourly vitals, lab draws, notes, resp events
snapshot_at_hour(h)	Cumulative state using only events where hour ≤ h
patient_view_at_hour()	Immutable PatientState view for agents & UI

In ML terms: this is the **serving-time feature view** with `as_of` timestamp semantics.

Every tab, chart, narrative, and brief recomputes from the snapshot — not the full stay.

Design choice: Safety-critical outputs are grounded and deterministic; LLM only enriches

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Airflow — Batch Temporal Reasoning

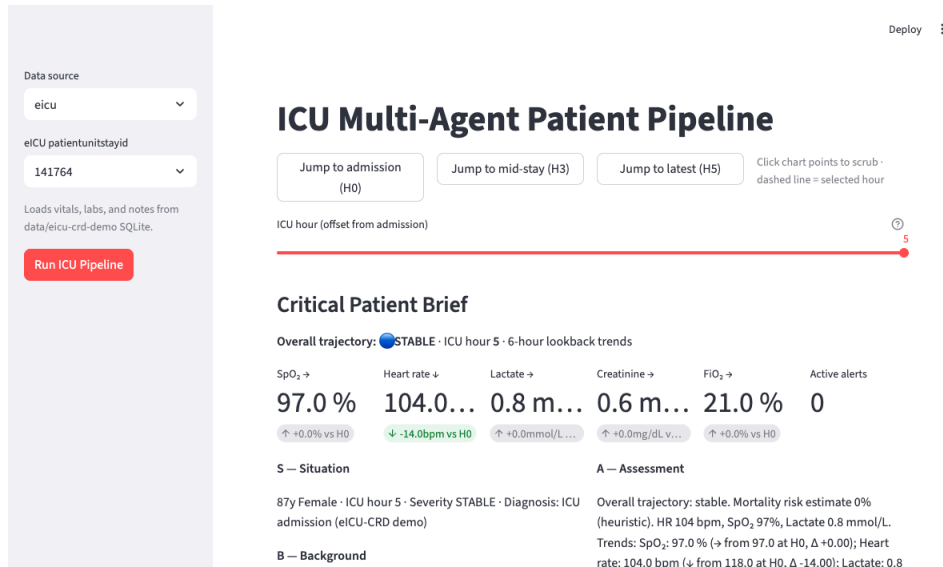
DAG: `icu_temporal_reasoning_pipeline`

1. Load eICU vitals/labs bundle
2. `build_hourly_reasoning_trace` → `stable` / `worsening` / `critical` per hour
3. Parallel branches: **hypoxia** ($\text{SpO}_2 < 85$) · **sepsis** (lactate > 4)
4. `coordinator_agent` → FiO_2 /PEEP adjustments, sepsis action list
5. Persist `reasoning_trace`, `temporal_narrative` → `output/timelines/`

DAG: `icu_multi_agent_pipeline` — load stay → risk → narrative → JSON + dashboard metadata

Streamlit = interactive single-stay inference · Airflow = scheduled, reproducible batch artifacts

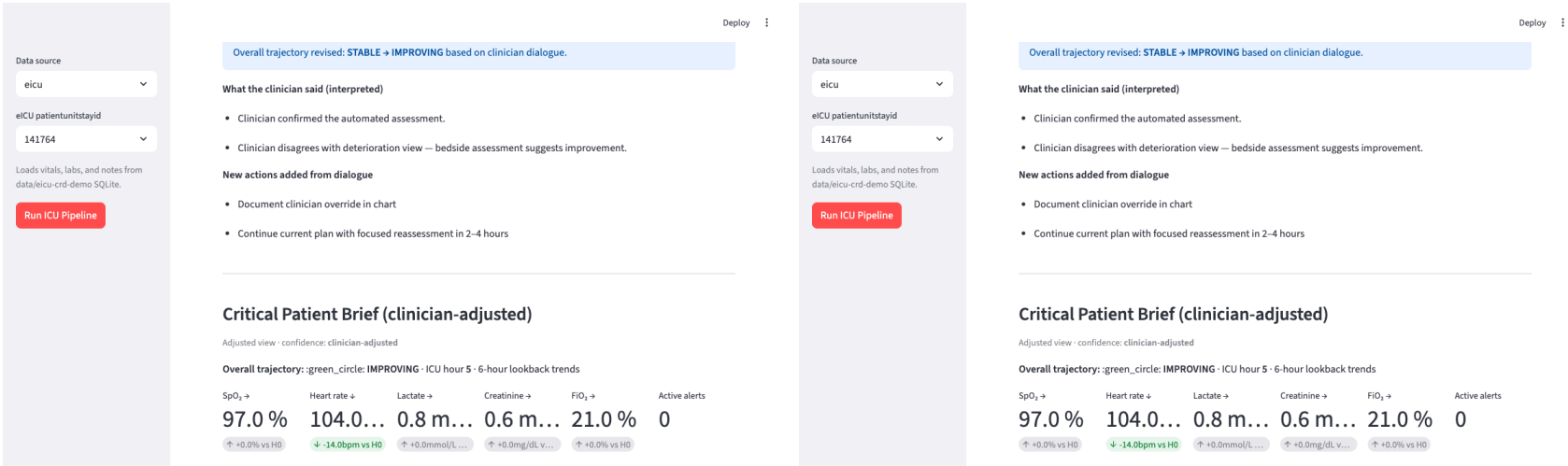
Critical Patient Brief



Automated decision-support panel at selected ICU hour

- Overall trajectory: STABLE / IMPROVING / DETERIORATING
- **Six-hour trend deltas** — SpO₂, HR, lactate, creatinine, FiO₂
- **Explainable alerts** with hour-stamped evidence (hypoxemia, tachycardia, AKI)

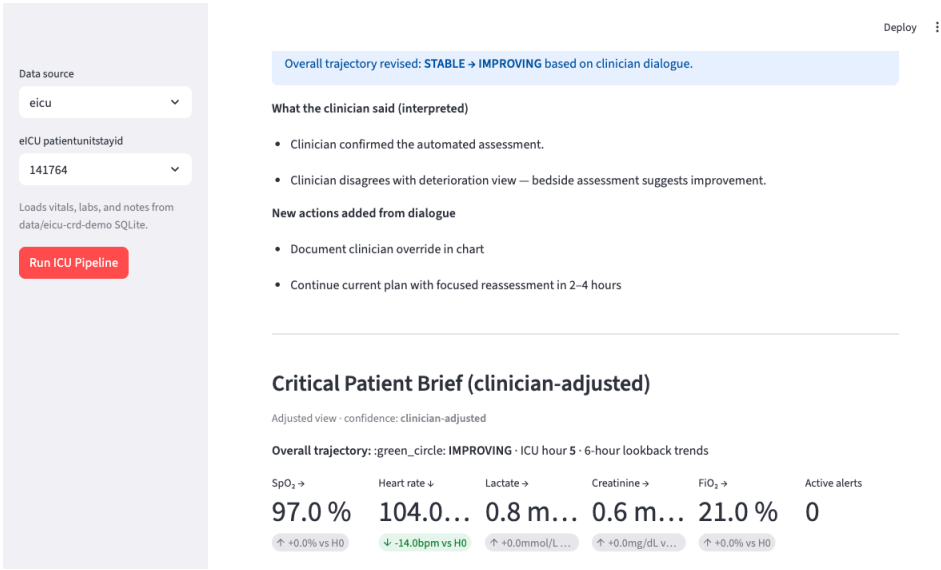
Streamlit Dashboard — Demo Walkthrough



Temporal navigation: hour slider · jump buttons · click-to-scrub on Plotly charts

Tab	Purpose
Overview	Consolidated bedside snapshot
Vitals	Multi-axis time-series through selected hour
Labs & Risk	Lab panel, mortality/sepsis/deterioration scores

Human-in-the-Loop — Feedback Overlay



Clinician: *"I disagree — the patient may be improving"*

Before	After
Trajectory: STABLE	Trajectory: IMPROVING
Automated alerts active	Matched alerts suppressed
Rule-based actions	Clinician-directed actions added

Evaluation & Limitations

Evaluated (systems correctness)

- Pydantic schema validity · eICU mapping fidelity
- Risk monotonicity (higher lactate / lower SpO₂ → higher score)
- Point-in-time integrity · temporal severity alignment
- Feedback overlay correctness · UI hour-scrub synchronization

Limitations (v1)

- Risk not calibrated to SOFA/APACHE · template narratives
- Sparse vitals in some demo stays · no clinician inter-rater study
- Dialogue audit logs not persisted · research prototype only

Conclusion & Future Work

Delivered: Grounded clinical ML pipeline — eICU ingestion, temporal snapshots, interpretable risk & briefs, optional LLM dialogue, HITL feedback overlay, Airflow orchestration

Future: SOFA/APACHE calibration · feedback audit store · learned severity models · LLM faithfulness eval · meds/treatment timeline · FHIR export · BeeAI deployment

Repository: github.com/suryashekhawat/MTech4SEM

Thank you

Questions? · Live demo available at `streamlit run ui/streamlit_app.py`